

SEMESTER END EXAMINATIONS – SEPTEMBER / OCTOBER 2023

Program	: B.E :- Common to CSE / ISE / CSE(CY) / AI & DS / BT / AI & ML / CSE (AI&ML)	Semester	: II
Course Name	: Engineering Chemistry	Max. Marks	: 100
Course Code	: CYC22	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- What is electrode potential? Derive the Nernst equation for single electrode potential. CO1 (08)
 - Explain the following battery characteristics: CO1 (06)
 - Energy density
 - Energy efficiency.
 - What are concentration cells? Two copper electrodes placed in copper sulphate solutions of equal concentration are connected to form a concentration cell. What is the cell voltage if one of the solutions is diluted until the concentration of Cu^{2+} ions is $1/5^{\text{th}}$ of its original value? What is the cell voltage after dilution? CO1 (06)
- What are reference electrodes? How to determine unknown electrode potential using calomel electrode. CO1 (08)
 - Write the cell representation, electrode reactions, and calculate the electrode potential at 298K for a galvanic cell formed by immersing Zinc rod in $5 \times 10^{-2}\text{M}$ zinc nitrate solution and silver rod in $1.2 \times 10^{-1}\text{M}$ silver nitrate solution. The standard E^0 values of Zn and Ag electrodes are -0.76V and +0.80V respectively. CO1 (06)
 - Write the construction and working of Lithium-ion battery. CO1 (06)

UNIT - II

- Outline: (i) Cathodic protection (ii) Waterline corrosion. CO2 (08)
 - Give reason: (i) Zinc gives better protection to iron than tin under corrosive environment. (ii). Wire mesh corrodes faster at the joints. (iii) Larger cathodic area results in intense corrosion. CO2 (06)
 - Explain the mechanism of electrochemical corrosion with an example. CO2 (06)
- What is differential aeration corrosion? Explain waterline pitting corrosion. CO2 (08)
 - What are corrosion inhibitors? Categorize inhibitors used for corrosion control giving two examples. CO2 (06)
 - Justify: CO2 (06)
 - Iron corrodes faster than aluminium, even though iron placed below aluminium in electrochemical series.
 - Zinc is more readily corroded when coupled with copper than with lead.

UNIT - III

- Write the preparation and uses of following polymers: CO5 (08)
 - Teflon
 - PMMA.
 - Mention the applications of liquid crystals in liquid crystal displays. CO5 (06)
 - Elaborate the Liquid crystalline behaviour in PAA series. CO5 (06)

6. a) What are conducting polymers? Give the conduction mechanism in polyacetylene. CO5 (08)
b) Define the terms: CO5 (06)
(i) Glass transition temperature (ii) Liquid crystal (iii) Polymer.
c) What are liquid crystals? How are they classified? Give examples. CO5 (06)

UNIT- IV

7. a) What are nanomaterials and explain the hydrothermal method for the synthesis of nanomaterials. CO4 (08)
b) Explain in brief about the determination of iron by using potentiometric sensors. CO4 (06)
c) What are top-down approaches of synthesis of nanomaterials? Mention nano materials characterization techniques used. CO4 (06)
8. a) Explain the solution combustion synthesis method of nanomaterials and discuss the applications of nano materials. CO4 (08)
b) Mention the application of pH-sensor and write in brief about determination of pH of the solution by using pH-sensor. CO4 (06)
c) How the estimation of copper is done by using colorimetric optical sensors? CO4 (06)

UNIT - V

9. a) Explain the effect of e-waste on human health. CO5 (08)
b) Explain the construction and working of PV cell. CO5 (06)
c) Elaborate on recycling of E-waste. CO5 (06)
10. a) How fuel cells differ from battery. Explain the construction and working of methanol-oxygen fuel cell. CO5 (08)
b) Explain the construction and working of polymer electrolyte fuel cell. CO5 (06)
c) Discuss in details the extraction of gold from e-waste. CO5 (06)
